

IP Addressing (Part 2)

The Binary Number System

Computers do not understand decimal numbers the way humans do. Everything inside a computer is stored as binary, a system that uses only two digits: 0 meaning OFF, and 1 meaning ON. Every IP address you see on a screen is actually a sequence of 0s and 1s underneath.

Understanding binary is what separates someone who can read IP addresses from someone who just copies them from a screen.

The 8-Bit Place Value Table

In binary, each position is called a bit. Each bit has a place value that doubles as you move left. Eight bits together make one octet. The place values across those 8 positions are fixed and always the same:

The 8-Bit Place Value Table

Each bit has a place value that doubles as you move left. Eight bits make one octet.

Bit Position (Left to Right)	Position 7	Position 6	Position 5	Position 4	Position 3	Position 2	Position 1	Position 0
Exponent	2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0
Power of 2	128	64	32	16	8	4	2	1
Decimal Value	128	64	32	16	8	4	2	1
Binary (if bit is 1)	1	1	1	1	1	1	1	1

Maximum Value for
One Octet (8 Bits)

$$128 + 64 + 32 + 16 + 8 + 4 + 2 + 1 = 255$$

1	1	1	1	1	1	1	1
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Binary: **11111111**
Decimal: **255**



When all 8 bits are 1, we add all the place values: $128 + 64 + 32 + 16 + 8 + 4 + 2 + 1 = 255$
This is why the highest possible value for one IPv4 octet is 255.

How to convert Binary to Decimal

The process is simple. You go through the place value table from left to right. For each column, ask one question: is my number greater than or equal to this column value? If yes, write a 1 and subtract. If no, write a 0 and move to the next column. Keep going until you have filled all 8 positions.

Think of it like making change. You start with the largest coin and work your way down.

Examples:

Here is the step-by-step conversion of 192.168.10.5 to binary.

Octet 1: 192

We repeatedly divide by 2 and track the remainders from bottom to top.

Division	Quotient	Remainder
$192 \div 2$	96	0
$96 \div 2$	48	0
$48 \div 2$	24	0
$24 \div 2$	12	0
$12 \div 2$	6	0
$6 \div 2$	3	0
$3 \div 2$	1	1
$1 \div 2$	0	1

Reading remainders from bottom to top: **11000000**

Octet 2: 168

Division	Quotient	Remainder
$168 \div 2$	84	0
$84 \div 2$	42	0
$42 \div 2$	21	0
$21 \div 2$	10	1
$10 \div 2$	5	0
$5 \div 2$	2	1
$2 \div 2$	1	0
$1 \div 2$	0	1

Reading remainders from bottom to top: **10101000**

Octet 3: 10

Division	Quotient	Remainder
$10 \div 2$	5	0
$5 \div 2$	2	1
$2 \div 2$	1	0
$1 \div 2$	0	1

We have only 4 bits, so we pad with leading zeros to make 8 bits: **00001010**

Octet 4: 5

Division	Quotient	Remainder
$5 \div 2$	2	1
$2 \div 2$	1	0
$1 \div 2$		1

We have only 3 bits so we pad with leading zeros to make 8 bits: **00000101**

Final Answer

Combining all four octets:

192.168.10.5 = **11000000.10101000.00001010.00000101**

Identifying IP Addresses: Public, Private and Class

You do not need to memorize a long list of rules to identify an IP address. You only need to look at the first octet. That single number tells you the class. The class tells you whether it is private. Everything else follows from there.

The Three Private Ranges You Must Know

Class	Private Range	First Octet	Common Use
A	10.0.0.0 to 10.255.255.255	Always 10	Large enterprises, Azure VNets, data centres
B	172.16.0.0 to 172.31.255.255	172 with second octet 16 to 31	Medium businesses and ISP internal networks
C	192.168.0.0 to 192.168.255.255	192 with second octet always 168	Home networks and small offices

Address Class Reference Table

Class	First Octet Range	Default Mask	Max Hosts per Network	Designed For
A	1 to 126	255.0.0.0 (/8)	16,777,214	Governments, ISPs, very large organizations
B	128 to 191	255.255.0.0 (/16)	65,534	Universities and large enterprises
C	192 to 223	255.255.255.0 (/24)	254	Small offices and home networks
D	224 to 239	N/A	N/A	Multicast groups only
E	240 to 255	N/A	N/A	Research and experimentation only

Quick Identification Hacks

Hack 1: The First Octet Rule

Look at the first octet only. If it is from 1 to 126 you are in Class A. If it is 128 to 191 you are in Class B. If it is 192 to 223 you are in Class C. Everything else is either special or reserved. You do not need to read the rest of the address to know the class.

Hack 2: Private Address Shortcuts

Starts with **10.** anything? Private Class A. Starts with **172.** and second octet is between 16 and 31? Private Class B. Starts with **192.168.**? Private Class C. Anything outside these three patterns is public.

Hack 3: Special Addresses to Memorize

127.0.0.1 is always loopback. Traffic goes back to the same machine. Never leaves. Any address starting with **169.254** is APIPA, which means DHCP failed on that device.

255.255.255.255 is a broadcast to the entire local network.

Hack 4: The Validity Check

Before you do anything with an IP address, scan each octet. If any single octet is above 255 the address is invalid and no further analysis is needed. You can stop right there.

Use the hacks above to work through each example below before reading the answer.

IP Address	Class	Public or Private	Type / Notes
192.168.1.1	C	Private	Typical home router gateway address
10.0.0.1	A	Private	Common in enterprise networks and Azure VNets
172.16.0.10	B	Private	Second octet is 16, which is within the private B range
172.32.0.1	B	Public	Second octet is 32, which is outside the private B range (16 to 31)
8.8.8.8	A	Public	Google DNS. First octet is 8, Class A range but not the private 10.x.x.x block
127.0.0.1	Special	Neither	Loopback address. Points back to the device itself. Never routed.
169.254.1.5	Special	Neither	APIPA address. Device could not reach a DHCP server. Investigate immediately.
203.0.113.5	C	Public	First octet is 203, Class C range but does not start with 192.168 so it is public
192.300.1.5	Invalid	N/A	Second octet is 300 which exceeds 255. This address cannot exist.
255.255.255.0	Special	Neither	Subnet mask notation, not a host address. You will use this constantly in subnetting.